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8. Consider a Library Management System. Develop a JavaScript program that will validate the controls in the forms you have created for the application. State the assumptions you make (business logic you are taking into consideration). Note: Your application must access a database using Servlet/JSP.

```
login.jsp X
src > main > webapp > WEB-INF > jsp > login.jsp > ? Review Next File
1 <%@ page contentType="text/html; charset=UTF-8" %>
2 <!DOCTYPE html>
3 <html>
4 <head>
5     <title>Admin Login</title>
6     <link rel="stylesheet" href="${pageContext.request.contextPath}/assets/css/style.css">
7     <script src="${pageContext.request.contextPath}/assets/js/validation.js" defer></script>
8 </head>
9 <body class="login-body">
10 <div class="login-card">
11     <h2>Library Management Login</h2>
12     <% if (request.getAttribute("error") != null) { %>
13     <div class="alert"><%= request.getAttribute("error") %></div>
14     <% } %>
15     <form method="post" action="${pageContext.request.contextPath}/login" onsubmit="return validateLoginForm(t
16         <label>Username</label>
17         <input type="text" name="username" required>
18         <label>Password</label>
19         <input type="password" name="password" required>
20         <button type="submit">Sign In</button>
21     </form>
22     <p class="hint">Default: admin / admin123</p>
23 </div>
24 </body>
25 </html>
26 |
```

```
dashboard.jsp X Browser Tab Browser Tab Browser Tab
src > main > webapp > WEB-INF > jsp > dashboard.jsp > ? ? ? ? ?
1 <%@ page contentType="text/html; charset=UTF-8" %>
2 <%@ taglib prefix="c" uri="jakarta.tags.core" %>
3 <%
4 <!-- HTML TYPE HTML -->
5 <html>
6 <head>
7     <title>Dashboard</title>
8     <link rel="stylesheet" href="${pageContext.request.contextPath}/assets/css/style.css">
9 </head>
10 <body>
11 <div class="layout">
12     <%@ include file="includes/sidebar.jspf" %>
13     <main class="content">
14         <%@ include file="includes/header.jspf" %>
15         <%@ include file="includes/message.jspf" %>
16         <section class="cards">
17             <div class="card"><h3>Total Books</h3><p>${stats.totalBooks}</p></div>
18             <div class="card"><h3>Total Members</h3><p>${stats.totalMembers}</p></div>
19             <div class="card"><h3>Issued Books</h3><p>${stats.issuedBooks}</p></div>
20             <div class="card"><h3>Available Books</h3><p>${stats.availableBooks}</p></div>
21         </section>
22         <section class="panel">
23             <h3>Recent Issued Books</h3>
24             <table>
25                 <thead><tr><th>ID</th><th>Book</th><th>Member</th><th>Issue Date</th><th>Due Date</th><th>Status</th></tr></thead>
26                 <tbody>
27                     <c:forEach var="item" items="${recentIssues}">
28                         <tr>
29                             <td>${item.id}</td>
30                             <td>${item.bookTitle}</td>
31                             <td>${item.memberName}</td>
32                             <td>${item.issueDate}</td>
33                             <td>${item.dueDate}</td>
34                             <td>${item.status}</td>
35                         </tr>
36                     </c:forEach>
37                 </tbody>
38             </table>
39         </section>
40         <section class="panel">
41             <h3>Top Categories</h3>
42             <div class="category-grid">
43                 <c:forEach var="cat" items="${topCategories}">
44                     <div class="category-item">
45                         <strong>${cat.category}</strong>
46                         <span>${cat.count} books</span>
47                     </div>
48                 </c:forEach>
49             </div>
50         </section>
51     </main>
52 </div>
53 </body>
54 </html>
```

[illegible]

```
<html>
<head>
<title>Books</title>
<link rel="stylesheet" href="${pageContext.request.contextPath}/assets/css/style.css">
<script src="${pageContext.request.contextPath}/assets/js/validation.js" defer></script>
</head>
<body>
<div class="layout">
<%% include file="includes/sidebar.jspf" %>
<main class="content">
<%% include file="includes/header.jspf" %>
<%% include file="includes/message.jspf" %>
<section class="panel">
<h3>[editBook != null ? 'Edit Book' : 'Add Book']</h3>
<form method="post" action="${pageContext.request.contextPath}/books" onsubmit="return validateBookForm(this);">
<input type="hidden" name="action" value="${editBook != null ? 'update' : 'add'}">
<c:if test="${editBook != null}">
<input type="hidden" name="id" value="${editBook.id}">
</c:if>
<div class="form-grid">
<input type="text" name="title" placeholder="Title" value="${editBook.title}" required>
<input type="text" name="author" placeholder="Author" value="${editBook.author}" required>
<input type="text" name="isbn" placeholder="ISBN" value="${editBook.isbn}" required>
<input type="text" name="category" placeholder="Category" value="${editBook.category}" required>
<input type="number" name="quantity" min="0" placeholder="Quantity" value="${editBook.quantity}" required>
</div>
<button type="submit">[editBook != null ? 'Update Book' : 'Save Book']</button>
</form>
</section>
<section class="panel">
<h3>View Books</h3>
<form class="toolbar" method="get" action="${pageContext.request.contextPath}/books">
<input type="text" name="q" value="${searchQuery}" placeholder="Search by title, author, ISBN, category">
<input type="number" name="size" value="${pageSize}" min="1">
<button type="submit">Search</button>
</form>
<table>
<thead><tr><th>ID</th><th>Title</th><th>Author</th><th>ISBN</th><th>Category</th><th>Quantity</th><th>Action</th></tr></thead>
<tbody>
<c:forEach var="book" items="${books}">
<tr>
<td>{book.id}</td><td>{book.title}</td><td>{book.author}</td><td>{book.isbn}</td><td>{book.category}</td><td>{book.quantity}</td>
<td>
<a class="btn-link" href="${pageContext.request.contextPath}/books/editId-{book.id}">Edit</a>
<form class="inline" method="post" action="${pageContext.request.contextPath}/books">
<input type="hidden" name="action" value="delete">
<input type="hidden" name="id" value="{book.id}">
<button class="danger" type="submit">Delete</button>
</form>
</td>
</tr>
</c:forEach>
</tbody>
</table>
</div>
</body>
</html>
```

```

src > main > resources > sql > library_db.sql
1 CREATE DATABASE IF NOT EXISTS library_db;
2 USE library_db;
3
4 DROP TABLE IF EXISTS issued_books;
5 DROP TABLE IF EXISTS members;
6 DROP TABLE IF EXISTS books;
7 DROP TABLE IF EXISTS admin;
8
9 CREATE TABLE admin (
10     id INT PRIMARY KEY AUTO_INCREMENT,
11     username VARCHAR(50) UNIQUE NOT NULL,
12     password VARCHAR(100) NOT NULL,
13     full_name VARCHAR(100) NOT NULL
14 );
15
16 CREATE TABLE books (
17     id INT PRIMARY KEY AUTO_INCREMENT,
18     title VARCHAR(255) NOT NULL,
19     author VARCHAR(255) NOT NULL,
20     isbn VARCHAR(20) UNIQUE NOT NULL,
21     category VARCHAR(100) NOT NULL,
22     quantity INT NOT NULL DEFAULT 0 CHECK (quantity >= 0)
23 );
24
25 CREATE TABLE members (
26     id INT PRIMARY KEY AUTO_INCREMENT,
27     name VARCHAR(100) NOT NULL,
28     email VARCHAR(100) UNIQUE NOT NULL,
29     phone VARCHAR(20) NOT NULL,
30     address VARCHAR(255) NOT NULL
31 );
32
33 CREATE TABLE issued_books (
34     id INT PRIMARY KEY AUTO_INCREMENT,
35     book_id INT NOT NULL,
36     member_id INT NOT NULL,
37     issue_date DATE NOT NULL,
38     due_date DATE NOT NULL,
39     return_date DATE NULL,
40     status VARCHAR(20) NOT NULL DEFAULT 'ISSUED',
41     FOREIGN KEY (book_id) REFERENCES books(id) ON DELETE RESTRICT,
42     FOREIGN KEY (member_id) REFERENCES members(id) ON DELETE RESTRICT
43 );
44
45 INSERT INTO admin (username, password, full_name) VALUES
46 ('admin', 'admin123', 'System Administrator');
47
48 INSERT INTO books (title, author, isbn, category, quantity) VALUES
49 ('Effective Java', 'Joshua Bloch', '9780134685991', 'Programming', 8),
50 ('Clean Code', 'Robert C. Martin', '9780132350884', 'Programming', 5),
51 ('The Pragmatic Programmer', 'Andrew Hunt', '9780201616224', 'Programming', 6),
52 ('Atomic Habits', 'James Clear', '9780735211292', 'Self Help', 10),
53 ('Deep Work', 'Cal Newport', '9781455586691', 'Productivity', 7),
54 ('Thinking Fast and Slow', 'Daniel Kahneman', '9780374533557', 'Psychology', 4);
55
56 INSERT INTO members (name, email, phone, address) VALUES
57 ('Aarav Mehta', 'aarav@example.com', '9876543210', 'Mumbai'),
58 ('Ishita Rao', 'ishita@example.com', '9876543211', 'Bengaluru'),
59 ('Rohan Singh', 'rohan@example.com', '9876543212', 'Delhi'),
60 ('Priya Sharma', 'priya@example.com', '9876543213', 'Pune');
61
62 INSERT INTO issued_books (book_id, member_id, issue_date, due_date, return_date, status) VALUES
63 (1, 1, CURDATE() - INTERVAL 3 DAY, CURDATE() + INTERVAL 11 DAY, NULL, 'ISSUED'),
64 (2, 2, CURDATE() - INTERVAL 5 DAY, CURDATE() + INTERVAL 9 DAY, NULL, 'ISSUED'),
65 (4, 3, CURDATE() - INTERVAL 18 DAY, CURDATE() - INTERVAL 4 DAY, CURDATE() - INTERVAL 2 DAY, 'RETURNED');
66

```

```

src > main > webapp > WEB-INF > jsp > members.jsp > ?
1 C<%@ page contentType="text/html; charset=UTF-8" %>
2 <%@ taglib prefix="c" uri="jakarta.tags.core" %>
3 <%
4
5
6 <!DOCTYPE html>
7 <html>
8 <head>
9     <title>Members</title>
10     <link rel="stylesheet" href="${pageContext.request.contextPath}/assets/css/style.css">
11     <script src="${pageContext.request.contextPath}/assets/js/validation.js" defer></script>
12 </head>
13 <body>
14 <div class="layout">
15     <%@ include file="includes/sidebar.jspf" %>
16     <main class="content">
17         <%@ include file="includes/header.jspf" %>
18         <%@ include file="includes/message.jspf" %>
19         <section class="panel">
20             <h3>${editMember != null ? 'Edit Member' : 'Add Member'}</h3>
21             <form method="post" action="${pageContext.request.contextPath}/members" onsubmit="return validateMemberForm(this);">
22                 <input type="hidden" name="action" value="${editMember != null ? 'update' : 'add'}">
23                 <c:if test="${editMember != null}">
24                     <input type="hidden" name="id" value="${editMember.id}">
25                 </c:if>
26                 <div class="form-grid">
27                     <input type="text" name="name" placeholder="Full Name" value="${editMember.name}" required>
28                     <input type="email" name="email" placeholder="Email" value="${editMember.email}" required>
29                     <input type="text" name="phone" placeholder="Phone" value="${editMember.phone}" required>
30                     <input type="text" name="address" placeholder="Address" value="${editMember.address}" required>
31                 </div>
32                 <button type="submit">${editMember != null ? 'Update Member' : 'Save Member'}</button>
33             </form>
34         </section>
35         <section class="panel">
36             <h3>View Members</h3>
37             <form class="toolbar" method="get" action="${pageContext.request.contextPath}/members">
38                 <input type="text" name="q" value="${searchQuery}" placeholder="Search by name, email, phone, address">
39                 <input type="number" name="size" value="${pageSize}" min="1">
40                 <button type="submit">Search</button>
41             </form>
42             <table>
43                 <thead><tr><th>ID</th><th>Name</th><th>Email</th><th>Phone</th><th>Address</th><th>Action</th></tr></thead>
44                 <tbody>
45                     <c:forEach var="member" items="${members}">
46                         <tr>
47                             <td>${member.id}</td><td>${member.name}</td><td>${member.email}</td><td>${member.phone}</td><td>${member.address}</td>
48                             <td>
49                                 <a class="btn-link" href="${pageContext.request.contextPath}/members?editId=${member.id}">Edit</a>
50                                 <form class="inline" method="post" action="${pageContext.request.contextPath}/members">
51                                     <input type="hidden" name="action" value="delete">
52                                     <input type="hidden" name="id" value="${member.id}">
53                                     <button class="danger" type="submit">Delete</button>
54                                 </form>
55                             </td>
56                         </tr>
57                     </c:forEach>

```

